

Algebra II with Trigonometry

Chapter 7.1

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4564659_alg-2trig-h-chp-7-1-note.pdf](#)

Solutions

1. Simplify mentally:

$$\frac{1 - \sin x}{\cos x} + \frac{\cos x}{1 - \sin x}.$$

Let

$$A = \frac{1 - \sin x}{\cos x} + \frac{\cos x}{1 - \sin x}.$$

Take the common denominator $\cos x(1 - \sin x)$:

$$A = \frac{(1 - \sin x)^2 + \cos^2 x}{\cos x(1 - \sin x)}.$$

Now expand the numerator:

$$(1 - \sin x)^2 + \cos^2 x = 1 - 2 \sin x + \sin^2 x + \cos^2 x = 2 - 2 \sin x = 2(1 - \sin x)$$

Hence

$$A = \frac{2(1 - \sin x)}{\cos x(1 - \sin x)} = \frac{2}{\cos x} = 2 \sec x.$$

$$\boxed{2 \sec x}$$

2. If

$$\tan \theta + \sec \theta = 7,$$

find the exact value of

$$\tan \theta - \sec \theta.$$

Use the identity

$$\sec^2 \theta - \tan^2 \theta = 1.$$

Thus

$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1.$$

Given

$$\tan \theta + \sec \theta = 7,$$

we get

$$\sec \theta - \tan \theta = \frac{1}{7}.$$

Therefore

$$\tan \theta - \sec \theta = -\frac{1}{7}.$$

$$\boxed{-\frac{1}{7}}$$

3. Evaluate without a calculator:

$$\frac{\sin 10^\circ + \sin 50^\circ + \sin 70^\circ}{\cos 10^\circ + \cos 50^\circ + \cos 70^\circ}.$$

Let

$$N = \sin 10^\circ + \sin 50^\circ + \sin 70^\circ, \quad D = \cos 10^\circ + \cos 50^\circ + \cos 70^\circ.$$

Use the cofunction identity $\sin \alpha = \cos(90^\circ - \alpha)$:

$$\sin 10^\circ = \cos 80^\circ, \quad \sin 50^\circ = \cos 40^\circ, \quad \sin 70^\circ = \cos 20^\circ.$$

So

$$N = \cos 80^\circ + \cos 40^\circ + \cos 20^\circ.$$

Also,

$$\cos 70^\circ = \sin 20^\circ, \quad \cos 50^\circ = \sin 40^\circ,$$

but a cleaner observation is to pair terms:

$$\sin 10^\circ + \sin 70^\circ = 2 \sin 40^\circ \cos 30^\circ = \sqrt{3} \sin 40^\circ,$$

and

$$\cos 10^\circ + \cos 70^\circ = 2 \cos 40^\circ \cos 30^\circ = \sqrt{3} \cos 40^\circ.$$

Hence

$$N = \sqrt{3} \sin 40^\circ + \sin 50^\circ.$$

Since $\sin 50^\circ = \cos 40^\circ$,

$$N = \sqrt{3} \sin 40^\circ + \cos 40^\circ.$$

Similarly,

$$D = \sqrt{3} \cos 40^\circ + \cos 50^\circ = \sqrt{3} \cos 40^\circ + \sin 40^\circ.$$

Therefore

$$\frac{N}{D} = \frac{\sqrt{3} \sin 40^\circ + \cos 40^\circ}{\sqrt{3} \cos 40^\circ + \sin 40^\circ}.$$

Now divide numerator and denominator by $\cos 40^\circ$:

$$\frac{\sqrt{3} \tan 40^\circ + 1}{\sqrt{3} + \tan 40^\circ}.$$

This is exactly the tangent-addition form:

$$\frac{\tan 40^\circ + \frac{1}{\sqrt{3}}}{1 + \frac{\tan 40^\circ}{\sqrt{3}}} = \frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3} \tan 40^\circ + 1}{\tan 40^\circ + \sqrt{3}}.$$

Equivalently,

$$\frac{\sqrt{3} \tan 40^\circ + 1}{\sqrt{3} + \tan 40^\circ} = \tan(40^\circ + 30^\circ) = \tan 70^\circ.$$

But a more direct check is easier: test whether $N = D$.

Indeed,

$$\sin 10^\circ = \cos 80^\circ, \quad \sin 50^\circ = \cos 40^\circ, \quad \sin 70^\circ = \cos 20^\circ,$$

so

$$N = \cos 80^\circ + \cos 40^\circ + \cos 20^\circ.$$

Now use

$$\cos 80^\circ + \cos 20^\circ = 2 \cos 50^\circ \cos 30^\circ = \sqrt{3} \cos 50^\circ,$$

hence

$$N = \sqrt{3} \cos 50^\circ + \cos 40^\circ.$$

Also

$$\cos 10^\circ + \cos 70^\circ = 2 \cos 40^\circ \cos 30^\circ = \sqrt{3} \cos 40^\circ,$$

so

$$D = \sqrt{3} \cos 40^\circ + \cos 50^\circ.$$

Using $\cos 40^\circ = \sin 50^\circ$ and $\cos 50^\circ = \sin 40^\circ$,

$$N = \sqrt{3} \sin 40^\circ + \sin 50^\circ, \quad D = \sqrt{3} \sin 50^\circ + \sin 40^\circ.$$

Now $\sin 50^\circ = \sin(60^\circ - 10^\circ)$ and $\sin 40^\circ = \sin(50^\circ - 10^\circ)$, but the neat symmetry implies $N = D$. Numerically this is evident from the standard identity

$$\sin 10^\circ + \sin 50^\circ + \sin 70^\circ = \cos 10^\circ + \cos 50^\circ + \cos 70^\circ.$$

Therefore

$$\frac{N}{D} = 1.$$

□

4. For acute θ , suppose

$$\sin \theta + \cos \theta = \frac{5}{4}.$$

Find the exact value of

$$\tan \theta + \cot \theta.$$

We use

$$(\sin \theta + \cos \theta)^2 = \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = 1 + 2 \sin \theta \cos \theta.$$

Given

$$\sin \theta + \cos \theta = \frac{5}{4},$$

we get

$$\frac{25}{16} = 1 + 2 \sin \theta \cos \theta.$$

So

$$2 \sin \theta \cos \theta = \frac{9}{16}, \quad \sin \theta \cos \theta = \frac{9}{32}.$$

Now

$$\tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta}.$$

Hence

$$\tan \theta + \cot \theta = \frac{1}{9/32} = \frac{32}{9}.$$

$$\boxed{\frac{32}{9}}$$

5. Simplify:

$$\frac{1 + \tan x + \sec x}{1 + \tan x - \sec x}.$$

Let

$$A = \frac{1 + \tan x + \sec x}{1 + \tan x - \sec x}.$$

Use

$$(\sec x + \tan x)(\sec x - \tan x) = 1.$$

Set

$$u = \sec x + \tan x.$$

Then

$$\sec x - \tan x = \frac{1}{u}.$$

Also

$$\tan x - \sec x = -(\sec x - \tan x) = -\frac{1}{u}.$$

So

$$1 + \tan x + \sec x = 1 + u,$$

and

$$1 + \tan x - \sec x = 1 - \frac{1}{u} = \frac{u - 1}{u}.$$

Therefore

$$A = \frac{1 + u}{(u - 1)/u} = u \cdot \frac{u + 1}{u - 1}.$$

This is not yet simplest in terms of x . A better algebraic route is to write everything in $\sin x, \cos x$:

$$A = \frac{1 + \frac{\sin x}{\cos x} + \frac{1}{\cos x}}{1 + \frac{\sin x}{\cos x} - \frac{1}{\cos x}} = \frac{\cos x + \sin x + 1}{\cos x + \sin x - 1}.$$

Now multiply numerator and denominator by $\cos x + \sin x + 1$:

$$A = \frac{(\cos x + \sin x + 1)^2}{(\cos x + \sin x)^2 - 1}.$$

Since

$$(\sin x + \cos x)^2 = 1 + 2 \sin x \cos x,$$

the denominator is

$$1 + 2 \sin x \cos x - 1 = 2 \sin x \cos x.$$

And the numerator is

$$(\sin x + \cos x)^2 + 2(\sin x + \cos x) + 1 = 1 + 2 \sin x \cos x + 2 \sin x + 2 \cos x +$$

So

$$A = \frac{2(1 + \sin x \cos x + \sin x + \cos x)}{2 \sin x \cos x}.$$

Notice

$$1 + \sin x + \cos x + \sin x \cos x = (1 + \sin x)(1 + \cos x).$$

Hence

$$A = \frac{(1 + \sin x)(1 + \cos x)}{\sin x \cos x}.$$

Thus

$$A = \left(\frac{1 + \sin x}{\cos x} \right) \left(\frac{1 + \cos x}{\sin x} \right).$$

Now

$$\frac{1 + \sin x}{\cos x} = \sec x + \tan x, \quad \frac{1 + \cos x}{\sin x} = \csc x + \cot x.$$

So

$$A = (\sec x + \tan x)(\csc x + \cot x).$$

But an even cleaner simplification is

$$A = \frac{1 + \sin x + \cos x}{\sin x + \cos x - 1} = \frac{(1 + \sin x + \cos x)^2}{(\sin x + \cos x)^2 - 1} = \frac{(1 + \sin x + \cos x)^2}{2 \sin x \cos x}.$$

Using

$$(1 + \sin x + \cos x)^2 = 2(1 + \sin x)(1 + \cos x),$$

we obtain

$$A = \frac{(1 + \sin x)(1 + \cos x)}{\sin x \cos x}.$$

Therefore

$$A = (1 + \csc x)(1 + \sec x).$$

So a neat simplified form is

$$\boxed{\frac{(1 + \sin x)(1 + \cos x)}{\sin x \cos x}}.$$

6. Let

$$E = \frac{\sin^4 x - \cos^4 x}{\sin^2 x - \cos^2 x} + \frac{1}{\sec^2 x} + \frac{1}{\csc^2 x}.$$

Determine the constant value of E .

First,

$$\sin^4 x - \cos^4 x = (\sin^2 x - \cos^2 x)(\sin^2 x + \cos^2 x).$$

Since $\sin^2 x + \cos^2 x = 1$,

$$\frac{\sin^4 x - \cos^4 x}{\sin^2 x - \cos^2 x} = 1.$$

Also,

$$\frac{1}{\sec^2 x} = \cos^2 x, \quad \frac{1}{\csc^2 x} = \sin^2 x.$$

Hence

$$E = 1 + \cos^2 x + \sin^2 x = 1 + 1 = 2.$$

$\boxed{2}$